

VANGSNES, Norway

WMO: 013380

Lat: 61.171N

Lon: 6.647E

Elev: 167

StdP: 14.61

Time Zone: 1.00 (EUC)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.2	22.2	7.7	8.2	25.1	10.4	9.4	25.8	25.4	38.0	23.2	37.8	9.5	80	0.623

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	75.4	61.7	71.6	59.8	68.4	58.5	63.4	72.1	61.7	69.2	60.2	66.7	7.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
59.8	77.2	66.7	58.2	72.8	65.5	56.8	69.1	64.0	28.8	72.2	27.6	69.4	26.6	66.8	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.6	18.3	16.4	19.2	79.8	3.3	4.6	16.8	83.1	14.9	85.8	13.0	88.4	10.6	91.7		
			16.8	65.0	3.3	2.2	14.4	66.6	12.5	67.9	10.7	69.2	8.3	70.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.2	33.9	34.1	37.3	43.9	51.2	57.0	61.3	59.4	53.5	46.1	39.6	35.8	(6)
	DBStd	11.04	5.46	5.66	5.08	5.13	6.12	4.73	5.24	4.20	4.40	4.89	5.43	6.02	(7)
	HDD50	2502	499	446	395	194	56	3	0	0	16	139	314	440	(8)
	HDD65	6915	964	866	860	632	429	244	141	181	345	586	762	905	(9)
	CDD50	1103	0	0	0	11	95	214	352	291	121	18	1	0	(10)
	CDD65	42	0	0	0	0	2	5	28	7	0	0	0	0	(11)
	CDH74	169	0	0	0	0	10	21	125	13	0	0	0	0	(12)
	CDH80	18	0	0	0	0	2	0	16	0	0	0	0	0	(13)
(14) Wind	WSAvg	8.1	9.8	8.7	8.4	7.2	7.6	8.1	7.9	7.2	7.2	7.9	8.4	9.0	(14)
(15) Precipitation	PrecAvg	51.3	5.9	5.0	4.7	2.6	2.2	2.7	3.2	3.5	4.7	5.5	5.3	6.3	(15)
	PrecMax	77.5	13.9	12.1	18.3	5.4	5.0	6.7	5.3	6.6	8.9	15.2	11.7	13.6	(16)
	PrecMin	26.6	0.0	0.0	0.0	0.6	0.4	0.4	1.2	0.9	1.0	1.2	0.2	0.1	(17)
	PrecStd	11.1	3.7	3.5	3.8	1.3	1.2	1.6	1.0	1.4	2.4	3.3	3.2	3.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.2	47.7	49.8	61.6	73.7	77.0	82.1	75.9	66.2	62.4	53.6	49.1	(19)
		MCWB	42.6	41.1	41.7	49.1	58.0	59.8	65.0	62.2	56.3	53.1	47.4	44.0	(20)
	2%	DB	44.9	45.2	47.3	57.9	68.3	72.1	77.8	71.0	63.5	57.4	51.1	47.0	(21)
		MCWB	40.5	40.3	41.8	47.6	55.1	57.5	63.6	60.4	55.2	51.1	45.8	42.2	(22)
	5%	DB	42.7	43.3	45.6	54.8	64.4	68.2	73.1	68.0	61.5	54.3	49.0	44.9	(23)
		MCWB	39.1	38.7	40.7	45.7	53.2	55.4	61.7	59.5	54.4	49.1	44.5	40.8	(24)
	10%	DB	40.7	41.4	43.8	51.6	60.6	64.8	69.8	65.9	59.7	52.2	46.8	43.1	(25)
		MCWB	37.4	37.3	39.2	43.7	51.2	54.3	60.4	58.7	53.8	47.0	43.1	39.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.6	43.4	44.8	51.1	60.1	61.1	66.8	64.0	58.7	54.3	49.1	45.8	(27)
		MCDB	46.3	46.0	47.2	59.4	71.7	73.8	77.8	72.5	62.9	58.5	51.6	47.9	(28)
	2%	WB	41.3	41.2	43.1	48.5	56.4	58.6	64.7	62.2	57.1	52.2	47.3	43.1	(29)
		MCDB	43.9	44.3	46.1	56.2	66.0	68.5	74.3	68.7	60.6	56.7	50.0	45.8	(30)
	5%	WB	39.5	39.5	41.1	46.6	54.0	57.2	62.9	61.0	55.8	50.2	45.0	41.4	(31)
		MCDB	42.1	42.2	44.5	52.9	62.3	64.9	71.4	66.3	59.5	53.9	47.9	43.9	(32)
	10%	WB	37.9	38.1	39.8	44.8	52.0	55.9	61.2	59.6	54.6	48.3	43.0	40.0	(33)
		MCDB	40.3	40.8	43.3	50.2	59.9	62.6	68.8	64.6	58.4	51.1	46.1	42.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	5.4	7.0	9.5	11.2	11.0	10.3	8.7	7.3	6.1	5.1	5.2	(35)
		MCDBR	8.1	7.2	8.3	13.6	17.1	17.4	16.4	12.1	9.3	8.2	8.4	7.8	(36)
	5% WB	MCWBR	6.0	4.9	5.5	6.6	7.8	7.6	6.4	5.6	5.0	5.2	5.8	5.9	(37)
		MCWBR	7.3	7.3	7.9	11.9	15.7	13.9	14.7	10.7	7.9	7.8	8.0	7.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.243	0.288	0.322	0.357	0.361	0.358	0.371	0.364	0.336	0.313	0.257	0.214	(40)
		taud	2.077	2.188	2.160	2.141	2.293	2.402	2.398	2.416	2.482	2.414	2.153	1.978	(41)
	Ebn at Noon	Ebn at Noon	151	217	249	262	272	276	268	261	249	211	144	103	(42)
		Edh at Noon	12	22	31	39	36	33	33	30	23	18	11	8	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	70	230	616	1172	1518	1537	1462	1121	698	321	101	45	(44)
	RadStd	RadStd	16	66	127	199	231	169	165	111	88	48	17	6	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page